

System Actors

HR Manager

The main person analyzing compensation fairness and making decisions about salary adjustments.

System Administrator

Handles configuration, system maintenance, and keeps an eye on how the application is running.

Epics and User Stories

Epic 1: Employee Compensation Analysis

User Story 1

As an HR Manager, I want to upload employee compensation data so I can check salary fairness across the company.

User Story 2

As an HR Manager, I want the system to automatically clean up employee data, so inconsistent or incomplete records get fixed before I analyze them.

User Story 3

As an HR Manager, I want the system to use machine learning to predict fair salaries, so I can compare what employees should earn to what they actually get.

Epic 2: Inequity Detection & Fairness Analysis

User Story 4

As an HR Manager, I want the system to spot employees whose salaries are way off from the predicted fair ones, so I can notice any possible inequities.

User Story 5

As an HR Manager, I want the system to calculate gender pay gaps, so I can see if there are compensation differences between demographic groups.

User Story 6

As an HR Manager, I want the system to show me fairness metrics and put them on dashboards, so I can quickly grasp overall compensation trends in the company.

User Story 7

As an HR Manager, I want the system to give employee-level explanations, so compensation decisions are clear and easy to justify.

Epic 3: Recommendation & Decision Support

User Story 8

As an HR Manager, I want the system to give fairness recommendations, so I can take steps to fix compensation gaps.

User Story 9

As an HR Manager, I want to run salary adjustment simulations with a set budget, so I can see how changes in compensation affect fairness.

User Story 10

As an HR Manager, I want to compare fairness metrics before and after simulations, so I can tell if my salary adjustments actually made things better.

Epic 4: System Administration & Management

User Story 11

As a System Administrator, I want to manage system settings and configurations, so the app stays secure and keeps running smoothly.

User Story 12

As a System Administrator, I want to monitor how well the system performs and its stability, so users always get reliable service.

5. Functional Requirements ID

Functional Requirement

FR1 The system lets HR users upload employee datasets in CSV or Excel formats.

FR2 The system automatically preprocesses and cleans uploaded data.

FR3 The system predicts fair salaries with a machine learning model.

FR4 The system flags employees with possible salary inequities.

FR5 The system calculates fairness metrics and gender pay gap percentages.

FR6 The system offers dashboard visualizations and reports.

FR7 The system generates salary adjustment recommendations.

FR8 The system simulates salary changes with a user-defined budget.

FR9 The system gives employee-level explanations for transparency.

FR10 The system lets administrators manage system settings.

6. Non-Functional Requirements ID

Non-Functional Requirement

NFR1 The system gives clear, understandable, and transparent salary predictions.

NFR2 The application supports interactive dashboard visualization.

NFR3 The system processes uploaded datasets quickly.

NFR4 The system keeps fairness and ethical AI principles front and center.

NFR5 The application is user-friendly for HR professionals.

NFR6 The system can scale and integrate with enterprise HR systems in the future.